

The Fix: Intrinsic Aspect-Ratio Attribute Mapping

How modern browsers bypassed the CSS override problem to reserve layout space.

⛔ Pre-2020: HTML Attributes Destroyed

```
index.html & style.css



img {
  width: 100%;
  height: auto;
}
```

🏗️ 1. HTML DOM tree generated.
Intrinsic attributes: 800×400

📄 2. CSS Box Model completely overrides HTML
width: 100% nullifies width=800
height: auto nullifies height=400

⚠️ 3. Aspect Ratio is OBLITERATED.
Browser renders a 0px high line.
(CLS is guaranteed upon load)

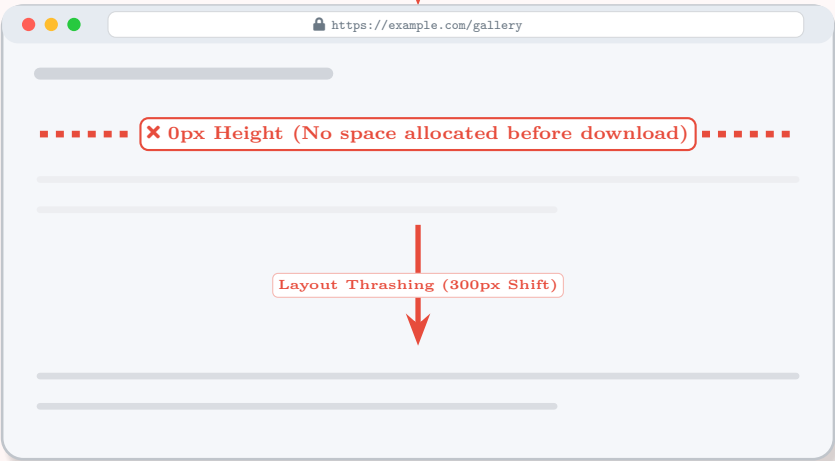


Image placeholder rendered at 0px height

⊗ Severe Layout Shift Guaranteed

Evolution

CSS Sizing 4
Spec Update

✅ Post-2020: The Silent UA Stylesheet Mapping

```
index.html & style.css

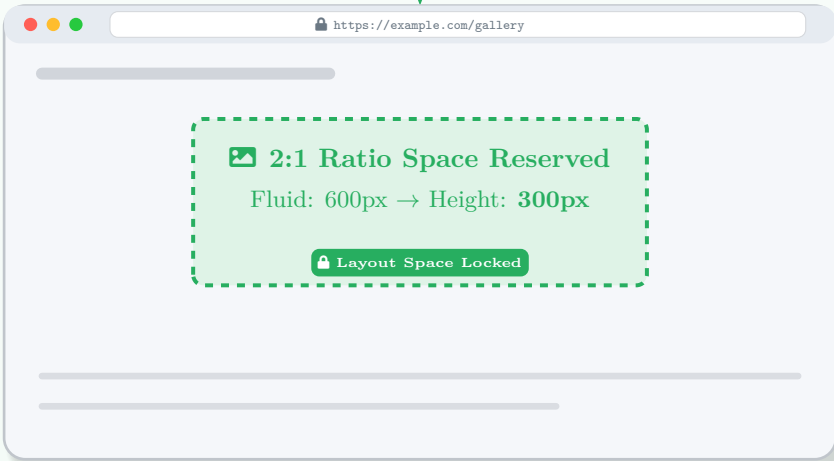


img {
  width: 100%;
  height: auto;
}
```

🏗️ 1. HTML DOM tree generated.
Intrinsic attributes: 800×400

🔧 2. User Agent Stylesheet maps a ratio:
aspect-ratio: attr(width) / attr(height)
aspect-ratio: 800 / 400 (Ratio 2:1)

🧮 3. CSS computes layout mathematically.
Fluid viewport width = 600px.
Height = 600px / 2 = 300px



Placeholder rendered perfectly at 300px height
BEFORE network request finishes

✅ Zero Cumulative Layout Shift (CLS = 0)